

RYAN BAIG

QUESTION 1: ESSAY

What is the meaning of rate-limiting step in a catalytic reaction. Explain in terms of rate constants and resistances.

Given Answer: The rate limiting step is responsible for how fast the catalytic reaction taking place will occur. For example if there is a reaction of adsorption, surface reaction, and desorption occurring. If the rate limiting step is the surface reaction, it will govern how fast the reaction takes place, because it is the slowest reaction. The other two reactions of adsorption and desorption would be implied as occurring much faster than the rate limiting step. The resistance for the rate limiting step is much greater than the other reactions taking place. Since the rate of reaction are equal, once we divide by the respective rate constant, the rate limiting step (r/K) is approximately zero. Note (resistance is $1/k$). The other rates divided by rate constant will be equal to one another and equal to zero from what was just stated.

Score: 25

QUESTION 2: ESSAY

List the seven steps of a catalytic reaction.

Given Answer:

1. mass transfer from the bulk to the surface
2. pore diffusion into the catalyst
3. adsorption of reactants
4. reaction (catalyst site)
5. desorption of the products
6. pore diffusion out of the catalyst
7. mass transfer out into the bulk

Correct Answer: 

Bulk Flow
Mass-Transfer in
Internal Diffusion in
Adsorption
Reaction
Desorption
Diffusion out
Mass-Transfer out
Bulk Flow out

Score: 25

QUESTION 3: ESSAY



For $A \rightarrow B$ reaction (Pressure = P_A) occurring on a catalyst surface, the combined rate of adsorption, desorption and surface reaction steps is given by $-r_A = (P_A^3 - P_B^2/5)/(1 + P_A + P_B)^2$. The mass transfer coefficient for A is given by k_c . When the system is agitated, the overall rate of the process also changes, implying that this is a mass-transfer limited process and the surface reaction is occurring very fast. Find the overall rate of reaction (at bulk pressures = P_A and P_B). Assume no internal diffusion limitation.

Given Answer: $r_C = (K_c/K_a) * (P_A^3 - P_B^2/5)/(1 + P_A + P_B)^2$

Score: 25

QUESTION 4: ESSAY



Why is it important for chemisorption to occur for catalysis? Why not physical adsorption?

Given Answer:

It is important because the atoms or molecules are held to the surface by valence forces of the same type as those that occur between bonded atoms in molecules. The result of this is that the electronic structure of the chemisorped molecule is perturbed significantly, which makes it significantly more reactive. This interaction with the catalyst causes bonds of the adsorbed reacted to be stretched, making them much easier to break. On the other hand, physical adsorption involves the forces of attraction between the gas molecules and the solid surface, which makes it significantly weaker. Due to the weak interactions by the van der Waals forces, only very small amounts can be adsorbed physically.

Score: 25